Shiv Nadar University, Chennai

Department of Information Technology

25IT1731 Computational Thinking and Problem Solving with Python

Exercise 1 – Data Types, Operators and Expressions

Part A

1. Priya wants to deposit ₹75,000 in a bank's fixed deposit scheme that offers a simple annual interest rate (R) of 7.5% for a period (T) of 3 years. Compute the Simple Interest (SI) she will earn at the end of 3 years and calculate the Maturity Amount (MA) she will receive.
    a. Identify the inputs required to solve the problem.
    b. Devise a solution and represent the same using flowchart.
    c. Develop a program to compute the simple interest and the maturity amount for the given deposit.

2. A civil engineering firm is building a mobile app to help site engineers quickly estimate the perimeter and area of construction plots of different shapes. The app records the dimensions relevant to each shape, the side of a square plot, the length and breadth of a rectangular plot, and the base and height of a triangular plot. Once the dimensions are entered, the app applies the appropriate formulas to compute the perimeter and/or area and displays the results.
    a. Identify the inputs required to solve the problem.
    b. Devise a solution and represent the same using algorithm.
    c. Develop a program to compute the perimeter and area of the given geometric shapes.

A hospital's neonatal unit records a newborn baby's body temperature as 37°C. For the hospital's international patient records and monitoring equipment (calibrated in Fahrenheit) and for research documentation (recorded in Kelvin), the temperature needs to be converted into both scales. Compute and display the equivalent temperature in Fahrenheit and Kelvin. Hint: *Fahrenheit = (Celsius × 9 / 5) + 32, Kelvin = Celsius + 273.15*

    a. Identify the inputs required to solve the problem.
    b. Devise a solution and represent the same using pseudocode.
    c. Develop a program to compute the temperature conversion.

3. Write a Python program to convert a given duration in seconds into hours, minutes and seconds. Use the floor division (//) and modulus (%) operators. For example, if the duration is 9875 seconds, the program should display the equivalent hours, minutes and seconds.

4. Write a Python program to calculate the distance and midpoint between two points (x1, y1) and (x2, y2). *Distance = sqrt((x2 − x1)² + (y2 − y1)²), Midpoint = ((x1 + x2) / 2 , (y1 + y2) / 2)*

5. An electrical engineering lab is analyzing an AC circuit where the voltage and current are represented as complex numbers, since they have both magnitude and phase. Given the voltage V = 8 + 6j volts and the current I = 3 − 4j amperes, the lab needs to compute the impedance-related quantities for their report. Develop a program to compute the sum, difference and product of V and I, and the magnitude of V, also print the real and imaginary parts of V (Hint: magnitude using abs())

6. A digital electronics lab is teaching students how permission flags are stored and combined at the bit level in operating systems. Each permission (Read, Write, Execute) is represented as a bit, and two permission sets are given as integers a = 22 and b = 13. Develop a program to compute the AND, OR, XOR, NOT, left shift and right shift of the given values, and use a bitwise operator to check whether a is an odd or even number. (Hint**:** a & 1 gives 0 if a is even and 1 if a is odd*)*

Part B

7. Write a Python program to calculate the kinetic energy, potential energy and total mechanical energy of an object. Consider an object of mass 10 kg, moving with a velocity of 12 m/s, at a height of 8 m above the ground. Take the acceleration due to gravity as 9.8 m/s². Calculate and display the kinetic energy, potential energy and total mechanical energy.

8. Write a program to convert lowercase characters to uppercase characters and print (without using build in functions).

9. A customer support chatbot for an e-commerce website needs to check incoming user messages for certain keywords, so it can route the query to the right department. A user sends the message: "I want to check the refund status of my order". The chatbot needs to check whether the word "refund" appears in the message, and also verify whether two variables referring to the department name actually point to the same string object in memory. (Hint: use in and is operator)

10. Write a Python program to find both roots of a quadratic equation $ax^2 + bx + c = 0$, for a = 2, b = 5, c = 7. (Hint: The roots may be complex, so use the cmath module)

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